CORKTOWN AUDIO WORKS

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BENCH REPORT

Service and function test

SKU	UNIT	BENCH DATE	TECHNICIAN
CAW-1090	Marantz 2220B receiver	2026-08-12	Kofi Mensah

INCOMING CONDITION

Brought up slowly on the dim-bulb tester: normal idle draw, no shorts, both channels playing. Dial lamps dim and patchy, meter lamp out. Volume, balance and tone pots scratchy; selector crackles between positions.

WORK PERFORMED

1. Dial and meter lamps replaced with new 8 V 300 mA fuse-style lamps (shop stock DL-8V300).
2. Pots and switches cleaned with DeoxIT D5 and worked through their full travel.
3. Bias set on both channels to the service manual value, rechecked after a 1 hour warm-up.
4. DC offset set at the speaker terminals on both channels (readings below).

MEASUREMENTS *(8 ohm resistive dummy load, 1 kHz sine, both channels driven)*

Test	Method	Left	Right	Target	Result
Output power	1 kHz, 12.65 V RMS into 8 ohms	20 W	20 W	20 W/ch into 8 ohms	Pass
THD	1 kHz at 20 W/ch	0.24%	0.27%	under 0.5%	Pass
DC offset	speaker terminals, after warm-up	7 mV	9 mV	under 10 mV	Pass

FUNCTIONAL CHECKS

Check	Notes	Result
FM tuner	Tunes smoothly end to end. Stereo lamp lights on WRCJ 90.9.	OK
AM tuner	Clean reception on WWJ 950, no buzz or whistle.	OK
Tuning meter	Needle swings and peaks on station.	OK
Phono, aux, tape monitor	Quiet and clean on both channels, volume well up.	OK
Volume, balance, bass, treble	No crackle or dropouts over the full travel.	OK
Speakers, headphone jack	Clean at both speaker outputs and the phones jack.	OK
Dial and meter lamps	All lit, even glow across the dial.	OK
Burn-in	2 hours at 1/3 power into the dummy load, no faults.	OK

COSMETIC NOTES

Faceplate: one fine hairline, visible only up close. No chips, dents or cracks. Knobs and dial glass clean.

RESULT

Signed off: Kofi Mensah, bench tech, 2026-08-12

PASS. All functions to spec.

20 W/ch into 8 ohms at under 0.3% THD, both channels. Tuning meter and stereo lamp OK.

Cosmetic grade is set from the master photos against Grading Guide v3. Ready for master photos.